Mode Reconstruction From IRs for Modal Synthesis

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Abstract

Modal synthesis is a physical modelling technique used to simulate the response of an object to excitatory signals by modelling the object as a system of resonators (modes) corresponding to the physical resonances of the object (Morrison and Adrien 1993). Conventionally, this can be done analytically or with physical simulations. Current known methods for doing this such as ESPRIT (Roy and Kailath 1989) can be quite slow for large impulse responses with some of the fastest methods described by Wells (2021) still taking time $O(n^2 \log(n))$. This project explores various potential methods that can do modal estimation in $O(mni)$ where m is the number of modes being found, n is the size of the impulse response in samples and i is the number of iterations for the algorithm.

1 Introduction

Modal synthesis is a physical modelling technique used to simulate the response of an object to excitatory signals by modelling the object as a system of resonators (modes) corresponding to the physical resonances of the object (Morrison and Adrien 1993). For example, if we want to reproduce the sound of a mallet hitting a xylophone, we can model the xylophone as a set of resonators and excite them with an impulse signal mimicking the impulse created by a mallet hitting a hard surface. In traditional modal synthesis, we pass the excitatory signal through a bank of resonant infinite impulse response (IIR) filters in parallel as shown in Figure 1. The resonant filters are described by second-order differential equations representing mass-spring systems and appear in the z-domain as holomorphic[1] functions with poles[2] near the unit circle corresponding to the resonant frequency of the corresponding mass-spring system (note that we are primarily interested in the z-domain because we will be working with discrete functions).

1.1 Impulse Responses and Convolution

Conventional convolution methods can be used to simulate complex linear systems using the impulse response[3] of the system. For example, the impulse response of a room can be used to create a reverb model that mimics the acoustic properties of the room to a high degree of accuracy. This is done by treating the impulse response as a finite impulse response (FIR) filter and passing a dry signal through the filter which is equivalent to taking the convolution of the dry signal with the IR (Valimaki et al. 2012).

Although this method can be very effective, it has a relatively high computational cost. In the time domain, this process takes $O(mn)$ where n is the length of our dry signal and m is the length of our IR. In the frequency domain, we can reduce this to $O(m \log(m))$ or $O(n \log(n))$ but this is often difficult to achieve in real-time scenarios. Most real-time convolution engines use a combination of time and

[1] A holomorphic function is a complex valued function that is analytic everywhere in its domain $D \subseteq \mathbb{C}$.

[2] For some holomorphic function H, poles are values of z for which $H(z) \to \infty$.

[3] The impulse response of a system is the response of a system to the Dirac delta function. In real life, it is sometimes impossible to get a perfect delta function but something like a clap can sometimes suffice.

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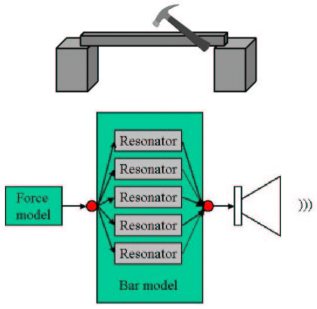


Figure 1: A modal synthesis block diagram showing an example of a bar being struck by a hammer (Doel and Pai 2007).

frequency domain techniques to leverage the real-time benefits of time domain processing and the reduced computation of frequency domain processing (Valimaki et al. 2012). This process requires a lot of precalculation which makes altering characteristics of the system over time impractical for real-time applications.

1.2 Project Goals

This project introduces and explores potential techniques for extracting modes from a given impulse response. While convolution reverb engines are very accurate, we can argue that determining a modal representation of an impulse response is useful because it gives us parametric control over the frequency response of the reverb that can be performed in real time. In a real-time scenario, the computational cost to calculate the response of a signal through a modal filter is $O(m)$ per sample where m is the number of modes in the filter. For a convolution engine which makes use of time domain operations, the computational cost for each sample is $O(n)$ where n is the length of the IR. In real-life applications, audio is often sampled at > 40kHz which means an IR that is only 1 second long results in $n > 40,000$ whereas, we may be able to summarize our IR with $m = 200$ modes[4]. This could lead to significant performance increases while still maintaining the tonal characteristics of the FIR associated with the convolution method. Approaches to modal synthesis such as the MOZAIC framework (Morrison and Adrien 1993) can only be used when the geometry of a space is known beforehand which isn't always easy to obtain. Methods for extracting modes from impulse responses such as ESPRIT algorithm by Roy and Kailath (1989) and the Modelled Pursuits (MoP) method proposed by Wells (2021) can be quite slow (the tests run by Wells 2021 use impulse responses of size 2000 because ESPRIT is quite intensive). MoP, which improves upon the run time of ESPRIT has a run time $O(n^2 \log(n))$.

In this project, we explore machine learning methods that can potentially be applied to this problem to decrease the runtime. We will be looking at regression using functions that approximate the shape of modes in the frequency domain and an expectation maximization algorithm that treats the frequency domain of the impulse response as a distribution. To simplify the parameter space, we will only look at matching the magnitude of the frequency response of a given impulse response. In theory, if we can find the centers of the modes and the decay rates, finding the phase of each mode can be done quite easily.

2 Explored Methodologies

All methods described were implemented in the Rust Programming Language since this project builds off of a previous project written in Rust.

[4]Note that both methods have runtimes with similar constant factors.

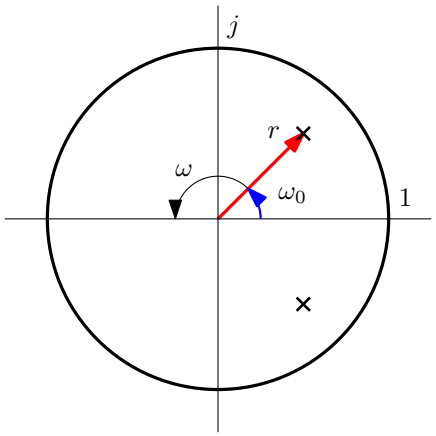


Figure 2: The poles controlled by Equation 1. Note that the magnitude of the frequency response is the magnitude of Equation 1 along the unit circle.

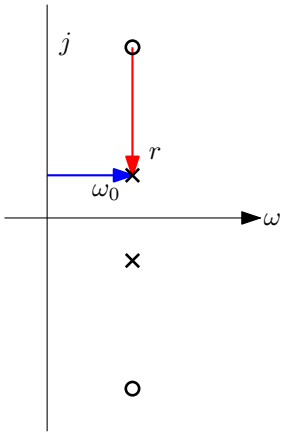


Figure 3: The poles controlled by Equation 3.

2.1 Regression Using Approximate Magnitude (Approximate Regression Method)

A resonant filter used in modal synthesis has an impulse response defined as an exponentially decaying sinusoid. Since we are ignoring phase for now, we can write it as:

$$h[n] = gr^n \sin(\omega_0 n)u[n]$$

where $g \in \mathbb{R}^+$ is some scale amount, $r \in (0, 1)$ is the rate of decay, ω_0 is the frequency of the sinusoid, and $u[n]$ is the unit step function. To get the frequency response, we take the z-transform of $h[n]$:

$$H(z) = \mathcal{Z}\{h[n]\} = \frac{gr\sin(\omega_0)z^{-1}}{(1 - re^{j\omega_0}z^{-1})(1 - re^{-j\omega_0}z^{-1})} = \frac{gr\sin(\omega_0)z^{-1}}{1 - 2r\cos(\omega_0)z^{-1} + r^2z^{-2}} \quad (1)$$

In the frequency domain, r corresponds to the magnitude of the two poles of $H(z)$. The decay rate of the exponential reflects how dampened a particular resonance is. As r approaches 1, the resonance at frequency ω decays much slower resulting in a stronger resonance. Using the inverse z-transform, we can derive the second order difference equation that can be used to determine the output of the system $y[n]$ to an incoming signal $x[n]$:

$$y[n] - 2r\cos(\omega_0)y[n-1] + r^2y[n-2] = gr\sin(\omega_0)x[n-1]$$
$$\implies y[n] = gr\sin(\omega_0)x[n-1] + 2r\cos(\omega_0)y[n-1] - r^2y[n-2]$$

This means that the response of a single filter to an incoming dry signal can be calculated in $O(n)$ where n is the length of the incoming dry signal since each sample can be calculated in constant time.

To simply things, as stated earlier, we will be working with the magnitude of the frequency domain to avoid additional complexity from imaginary numbers. Finding the magnitude of the frequency response requires substituting z with the parametrization of the unit circle in the complex plane $e^{j\omega}$:

$$|H(e^{j\omega})| = \frac{|gr\sin(\omega_0)e^{-j\omega}|}{|1 - 2r\cos(\omega_0)e^{-j\omega} + r^2e^{-2j\omega}|}$$
$$= \frac{gr\sin(\omega_0)}{\sqrt{(\cos(2\omega) - 2r\cos(\omega_0)\cos(\omega) + r^2)^2 + (\sin(2\omega) - 2r\cos(\omega_0)\sin(\omega))^2}} \quad (2)$$

This equation is quite complicated in terms of r which makes it messy to work with if we are trying to perform regression with gradient descent.

To simplify this, the Approximate Regression Method uses the fact that Equation 1 is holomorphic which means we can use the relative distances of poles and zeros in Equation 1 to create a real-valued function that mimics the shape of Equation 2. Equation 1 has poles that moves closer or farther from

the unit circle at frequencies corresponding to ω_0 depending on the value of r (Figure 2). We can mimic this behavior on the real number line by placing poles closer or farther to the real number line depending on the value of r (Figure 3). Doing this, we derive the equation:

$$M(\omega, g, r) = \frac{gr((\omega - \omega_0)^2 + 1)}{(\omega - \omega_0)^2 + (1 - r)^2} \tag{3}$$

The shape of Equation 3 has a very similar shape to Equation 2 on the domain from 0 to π which means we can approximate regression with Equation 2 by using Equation 3.

To perform regression, the Levenberg-Marquardt Algorithm (LMA) (Levenberg 1944) for least squares optimization is used. This means the function being minimized looks like:

$$f(\mathbf{x}, \mathbf{y}) = \sum_{i=1}^{n} \frac{1}{2} R(x^{(i)}, y^{(i)}, \mathbf{g}, \mathbf{r})^2 \tag{4}$$

$$R(x, y, \mathbf{g}, \mathbf{r}) = \sum_{j=1}^{m} \left[M(x, g^{(j)}, r^{(j)}) \right] - y \tag{5}$$

For LMA, we only need the derivative of R in terms of g and r which are straight forward to calculate. The overall algorithm takes $O(mn)$ to calculate the Jacobian matrix of the residual functions and each value of ω in the frequency domain which means it takes $O(mni)$ to run i iterations.

2.2 Approximate Expectation Maximization

Another approach that can potentially be used to convert IRs into modal spaces is expectation maximization where we treat the magnitude of the frequency response of the IR as a probability distribution. For simplicity, a Gaussian Mixture Model is used to estimate the center frequencies of modes along with their relative strength (larger variance indicates a faster decay rate since the mode is wider which can only happen if we decrease the value of r). However, there are a few issues with this approach that will be covered in the discussion section.

Instead of using points sampled from a distribution like a typical GMM, the EM algorithm that is implemented uses the magnitude corresponding to each frequency bin as a weight for each responsibility associated with the bin. This gives us the following EM equations:

$$r_j^{(i)} = p(z = j | x^{(i)}) = \frac{\pi_j \mathcal{N}\left(x^{(i)} | \mu_j, \sigma_j^2\right)}{\sum_{j'=1}^{m} \pi_{j'} \mathcal{N}\left(x^{(i)} | \mu_{j'}, \sigma_{j'}^2\right)} \tag{6}$$

$$\pi_j = \frac{1}{\sum_{i=1}^{n} y^{(i)}} \sum_{i=1}^{n} y^{(i)} r_j^{(i)} \tag{7}$$

$$\mu_j = \frac{1}{\sum_{i=1}^{n} y^{(i)} r_j^{(i)}} \sum_{i=1}^{n} y^{(i)} r_j^{(i)} x^{(i)} \tag{8}$$

$$\sigma_j^2 = \frac{1}{\sum_{i=1}^{n} y^{(i)} r_j^{(i)}} \sum_{i=1}^{n} y^{(i)} r_j^{(i)} (x^{(i)} - \mu_j)^2 \tag{9}$$

The implementation uses dynamic programming to prevent repeated calculations which means that each iteration of EM only takes $O(mn)$ since there are n weighted responsibilities for m distributions. The total run time of the algorithm is therefore $O(mni)$ where i is the number of iterations of EM.

One of the main potential drawbacks of EM is that we may not converge to a good minimum if we have a poor initialization. According to Chen (2013), we can improve the chances of converging to a good minimum by initializing our Gaussians at the means outputted by K-means or K-medioids. In our algorithm, we will initialize them at promising peaks with high prominence[5].

After obtaining a set of distributions that match the magnitude distribution, we can use something like regression to convert each Gaussian into modal data.

[5]By prominence, we mean topographic prominence.

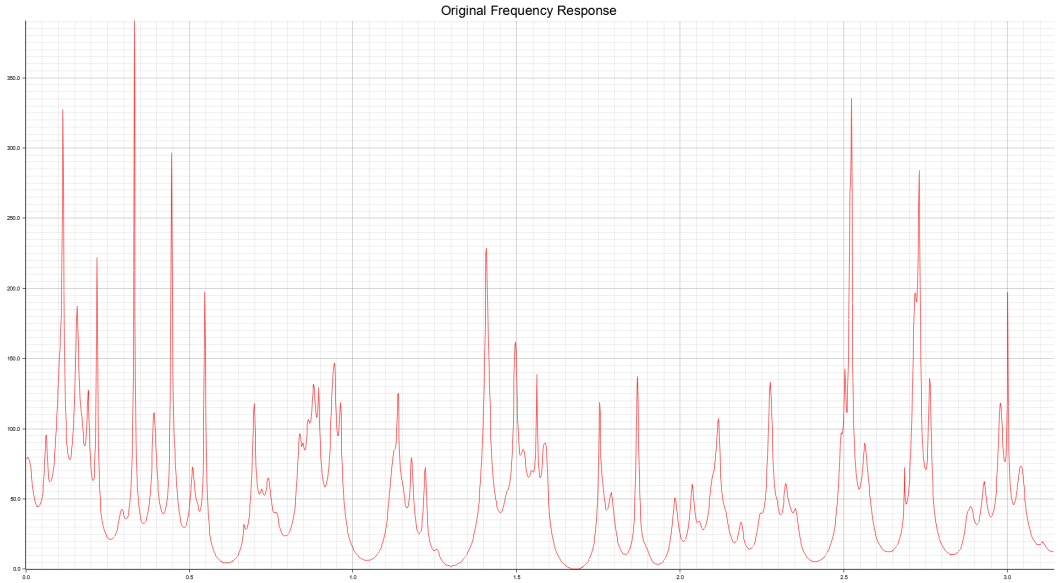


Figure 4: The frequency response of a set of 100 randomly generated modes.

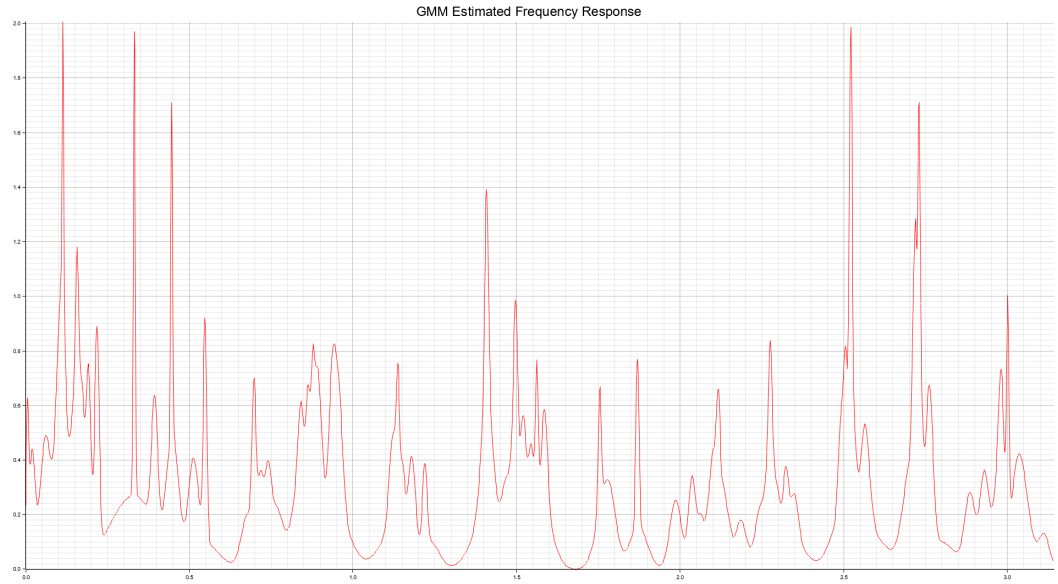


Figure 5: The fit achieved by the Approximate Expectation Maximization method on the frequency domain data presented in Figure 4.

3 Analysis and Discussion

While the Approximate Regression Method seems promising, the LMA library used made it difficult to use techniques such as barriers to try and prevent r and g from going outside of their expected range. This meant that for complex signals with many potential modes, LMA would often end up in local minima with r and g outside of their expected ranges. Because of this, there are no tested results associated with this method that are presented in this paper. Given more time, a better optimization algorithm could potentially be used to make this method more viable. The idea of representing the shape of the magnitude of the frequency response with a simpler function that approximates the shape seems promising and is a concept that is shared across both methods explored in this project.

The Approximate Expectation Maximization algorithm appears to work quite well. An experiment was conducted with 100 randomly generated modes, the frequency response of which can be seen in Figure 4. After running EM, we get the estimated distribution shown in Figure 5 which closely resembles the original (up to scaling since the output of EM is a proper distribution that sums to 1). As EM often converges faster than gradient descent, this algorithm is likely to use less iterations than other gradient descent methods.

One of the major drawbacks of the current EM algorithm is that Equation 2 tends to have a much larger magnitude further away from the center than Gaussian distributions. This means that for signals with very few modes, the fitted Gaussians tend to have a very large spread since they have to account for the points on the graph associated with a mode that are far away but still have significant magnitude. Given more time, there are a few potential techniques that could be used to mitigate this as well as improve the accuracy of the algorithm overall:

- We could come up with a new distribution that can be described similarly to Equation 3 which shares a similar shape to Equation 2 over the domain from 0 to π. This would mean we are fitting distributions that are more similarly shaped than Gaussians to the actual modes we are looking for and would prevent the issue that occurs with sparse modes. One potential way to do this could be creating a distribution that is a union of multiple Gaussians with parameters that are weighted proportionally. This could be simpler than coming up with a completely different distribution since we can use similar equations for calculating responsibility and updating the distributions.
- We could apply some kind of transform to the frequency response that makes the modes appear more Gaussian-like. It's a little less obvious how this could be done but it's still a potential approach.

Another important note is that both of these methods use approximation which may be less accurate than algorithms like ESPRIT (Roy and Kailath 1989) and MoP (Wells 2021) which use classical signal processing with mathematical guarantees.

4 Conclusion and Future Work

We have explored two different methods for determining modal representation from impulse response signals. While there is still work to be done on improving these techniques, they each have the potential to decrease the run time on current known methods (although they do sacrifice some accuracy). Given more time, it would be a good idea to explore different variants of the Approximate EM method since there is a clear path forward for how it can be improved upon using unions of Gaussian distributions instead of pure Gaussians. The code for everything in this project can be found at `https://github.com/Amarantheum/GPResonatorLibrary/tree/cpsc440/resonator_builder`.

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